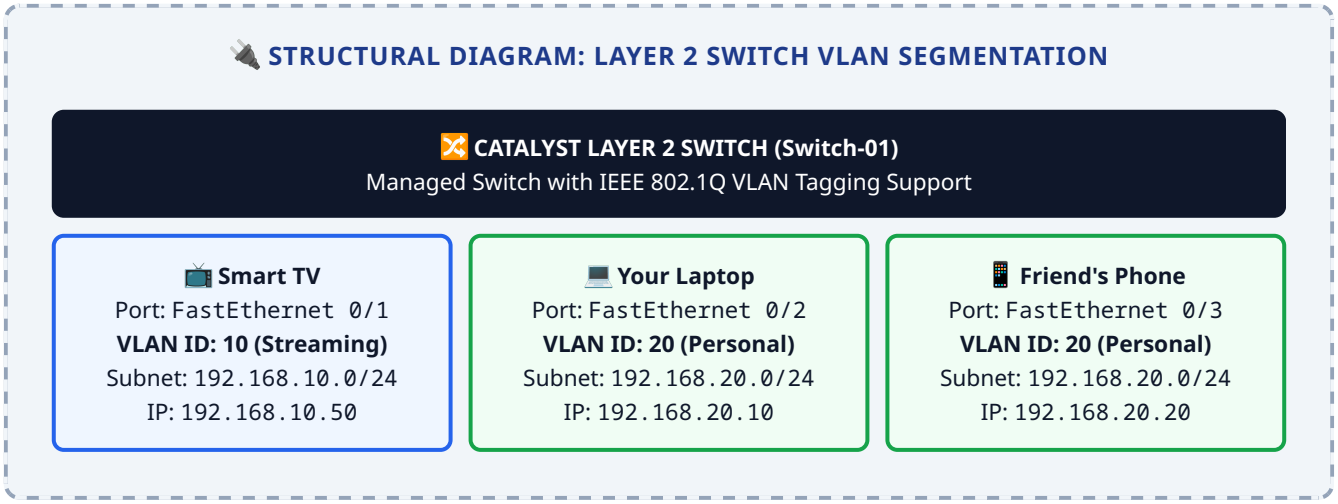
# Virtual LANs (VLANs) & Layer 2 Switching Assignment

Comprehensive Long-Form Technical Report & Practical Solutions (Tasks 1 to 4)

## Task 1: VLAN Network Architecture Diagram & Device Segmentation

A **Virtual Local Area Network (VLAN)** partitions a physical Layer 2 switch into multiple distinct logical broadcast domains. By logically isolating network traffic at Layer 2, devices in different VLANs cannot communicate with each other directly without a Layer 3 router, enhancing security, reducing broadcast storm traffic, and improving network management.




### Logical Isolation Mechanics:

- **VLAN 10 (Streaming Network):** Port Fa0/1 is assigned to VLAN 10. The Smart TV handles high-bandwidth 4K video traffic isolated from personal devices.
- **VLAN 20 (Personal Devices Network):** Ports Fa0/2 and Fa0/3 are assigned to VLAN 20. Your Laptop and your Friend's Phone can communicate directly with each other at Layer 2 because they share the same VLAN and broadcast domain.
- **Inter-VLAN Isolation:** Even though the Smart TV (VLAN 10) and your Laptop (VLAN 20) are plugged into the same physical switch, the switch drops frames attempting to cross VLAN boundaries.

## Task 2: Cisco Packet Tracer VLAN Configuration & Ping Isolation Verification

This simulation configures two distinct VLANs on a Cisco Catalyst 2960 switch in Cisco Packet Tracer: **VLAN 10 (Sales)** with 3 PCs and **VLAN 20 (HR)** with 2 PCs.

### Step 1: Switch CLI Configuration (VLAN Creation & Port Assignment)

```
! --- CISCO CATALYST SWITCH CLI CONFIGURATION ---
Switch# configure terminal
Switch(config)# hostname Switch-Core
```

```
! Create VLAN 10 and VLAN 20
Switch-Core(config)# vlan 10
Switch-Core(config-vlan)# name Sales
Switch-Core(config-vlan)# exit

Switch-Core(config)# vlan 20
Switch-Core(config-vlan)# name HR
Switch-Core(config-vlan)# exit

! Assign FastEthernet Ports 0/1 to 0/3 to VLAN 10 (Sales)
Switch-Core(config)# interface range fastEthernet 0/1 - 3
Switch-Core(config-if-range)# switchport mode access
Switch-Core(config-if-range)# switchport access vlan 10
Switch-Core(config-if-range)# exit

! Assign FastEthernet Ports 0/4 to 0/5 to VLAN 20 (HR)
Switch-Core(config)# interface range fastEthernet 0/4 - 5
Switch-Core(config-if-range)# switchport mode access
Switch-Core(config-if-range)# switchport access vlan 20
Switch-Core(config-if-range)# end
Switch-Core# write memory
```

### Step 2: End-Host IP Addressing & Verification Layout

| Device Name | Switch Port | Assigned VLAN | IP Address | Subnet Mask |
|---|---|---|---|---|
| **PC-1** | FastEthernet 0/1 | VLAN 10 (Sales) | 192.168.10.11 | 255.255.255.0 |
| **PC-2** | FastEthernet 0/2 | VLAN 10 (Sales) | 192.168.10.12 | 255.255.255.0 |
| **PC-3** | FastEthernet 0/3 | VLAN 10 (Sales) | 192.168.10.13 | 255.255.255.0 |
| **PC-4** | FastEthernet 0/4 | VLAN 20 (HR) | 192.168.20.14 | 255.255.255.0 |
| **PC-5** | FastEthernet 0/5 | VLAN 20 (HR) | 192.168.20.15 | 255.255.255.0 |

### Step 3: Connectivity Verification Results (Ping Tests)

**Test A: Intra-VLAN Ping (PC-1 to PC-2 within VLAN 10)**

```
PC-1> ping 192.168.10.12
Pinging 192.168.10.12 with 32 bytes of data:
Reply from 192.168.10.12: bytes=32 time<1ms TTL=128
Reply from 192.168.10.12: bytes=32 time<1ms TTL=128
Packet Result: SUCCESS (4/4 Received, 0% Loss)
```

**Test B: Inter-VLAN Ping (PC-1 in VLAN 10 to PC-4 in VLAN 20)**

```
PC-1> ping 192.168.20.14
Pinging 192.168.20.14 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Packet Result: FAILED (0/4 Received, 100% Loss)
```

> **Technical Explanation of Ping Result:**
> The ping between PC-1 and PC-4 fails because they reside in separate VLANs (VLAN 10 vs VLAN 20). The switch

> blocks frame forwarding between different VLAN IDs at Layer 2. Without a Layer 3 device (such as a Router performing Router-on-a-Stick or a Layer 3 Switch), cross-VLAN communication is completely blocked.

## Task 3: Reassigning a PC between VLANs (Before/After CLI Commands & Testing)

**Scenario:** PC-1 (connected to port `FastEthernet 0/1`) is initially in **VLAN 10**. We will reassign port `FastEthernet 0/1` to **VLAN 20** and change its IP address to match the VLAN 20 subnet, proving the transfer of membership.

### 1. BEFORE Reassignment (Initial State):

- Port Fa0/1 assigned to: **VLAN 10**
- PC-1 IP Address: 192.168.10.11/24
- **Connectivity Test:** PC-1 can successfully ping PC-2 (192.168.10.12 in VLAN 10), but cannot ping PC-4 (192.168.20.14 in VLAN 20).

### 2. CLI Reassignment Commands Executed on Switch:

```
! --- REASSIGNING PORT FA0/1 FROM VLAN 10 TO VLAN 20 ---
Switch-Core# configure terminal
Switch-Core(config)# interface fastEthernet 0/1
Switch-Core(config-if)# switchport access vlan 20
Switch-Core(config-if)# end

! Verify updated VLAN database assignment
Switch-Core# show vlan brief
```

### 3. AFTER Reassignment (Updated State & Host Reconfiguration):

We update PC-1's IP address to match the VLAN 20 subnet: 192.168.20.11/24.

```
! Test 1: PC-1 (now in VLAN 20) attempts to ping PC-2 (still in VLAN 10)
PC-1> ping 192.168.10.12
Request timed out. (FAILED - Correct behavior as they are now in different VLANs)

! Test 2: PC-1 (now in VLAN 20) pings PC-4 (VLAN 20)
PC-1> ping 192.168.20.14
Reply from 192.168.20.14: bytes=32 time<1ms TTL=128
Reply from 192.168.20.14: bytes=32 time<1ms TTL=128
Packet Result: SUCCESS (4/4 Received - Confirms PC-1 is now active in VLAN 20!)
```

## Task 4: Automated Multi-Port Batch Assignment (Range Command & Automation Script)

Assigning switchports individually is inefficient and prone to human error. Network operating systems provide range syntaxes and scripting to automate bulk port configuration.

### Option A: Cisco IOS Bulk Range Syntax (Cisco Packet Tracer / GNS3)

Cisco IOS utilizes the `interface range` command to configure contiguous or non-contiguous groups of ports in a single operation:

```
! --- CISCO IOS AUTOMATED BATCH PORT ASSIGNMENT ---
Switch-Core# configure terminal

! Create VLAN 30 if it does not already exist
Switch-Core(config)# vlan 30
Switch-Core(config-vlan)# name Guest-Wi-Fi
Switch-Core(config-vlan)# exit

! Select 10 ports simultaneously using interface range
Switch-Core(config)# interface range fastEthernet 0/10 - 19
Switch-Core(config-if-range)# description --- AUTOMATED GUEST VLAN 30 PORTS ---
Switch-Core(config-if-range)# switchport mode access
Switch-Core(config-if-range)# switchport access vlan 30
Switch-Core(config-if-range)# no shutdown
Switch-Core(config-if-range)# end

! Verify batch assignment across all 10 ports
Switch-Core# show vlan id 30
```

### Option B: Automated Python Script using Netmiko (For Enterprise Networks)

For automating across multiple switches programmatically without manual CLI entry:

```
# Python Script to automate VLAN 30 assignment on 10 ports using Netmiko
from netmiko import ConnectHandler

cisco_switch = {
    'device_type': 'cisco_ios',
    'host': '192.168.1.254',
    'username': 'admin',
    'password': 'SecurePassword123',
}

config_commands = [
    'vlan 30',
    'name Guest-Wi-Fi',
    'interface range fastEthernet 0/10 - 19',
    'switchport mode access',
    'switchport access vlan 30',
    'no shutdown'
]

net_connect = ConnectHandler(**cisco_switch)
output = net_connect.send_config_set(config_commands)
print("[+] Configured 10 ports successfully:
", output)
net_connect.disconnect()
```